

Introduction to Gaussian Processes

Presenter: Yan Peng

Mentor: Yuhan Qian

Introduction to Gaussian Processes



Introduction to Gaussian Processes



Linear Regression

- View of Optimization:

- MSE: $\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2.$

Linear Regression

- View of Optimization:

- MSE: $\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2.$



$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Linear Regression

- View of Optimization:

- MSE: $\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2.$



$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Pros:

- Computationally efficient
- Clear interpretation

Cons:

- Ineffective for capturing complex relationships.
- Potential lack of solutions
- No uncertainty quantification

Regularization

- Ridge regression:

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2.$$

Regularization

- Ridge regression:

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2. \quad \longrightarrow \quad \hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_2^2$$

for some pre-defined hyper-parameter $\lambda > 0$.

Regularization

- Ridge regression:

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2. \quad \longrightarrow \quad \hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_2^2$$

for some pre-defined hyper-parameter $\lambda > 0$.

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda I)^{-1} \mathbf{X}^\top \mathbf{y}.$$

Regularization

- Ridge regression: $\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda I)^{-1} \mathbf{X}^\top \mathbf{y}.$

Pros:

- **Guaranteed Solution:** Ridge regression always provides a solution.
- **Improved Generalization**

Cons:

- **Bias in Estimates**
- **Reduced Training Accuracy**

Introduction to Gaussian Processes



Introduction to Gaussian Processes



Bayesian Regression

- Standard linear regression model with Gaussian noise:

$$f(\mathbf{x}) = \mathbf{x}^\top \beta, \quad y = f(\mathbf{x}) + \epsilon. \quad \epsilon \sim N(0, \sigma_n^2)$$

- Gaussian Prior on β :

$$\beta \sim N(0, \Sigma_p)$$

Bayesian Regression

- Posterior distribution:

$$p(\beta \mid \mathbf{X}, \mathbf{y}) \propto \exp\left\{-\frac{1}{2}(\beta - \hat{\beta})^\top \left(\frac{1}{\sigma_n^2} \mathbf{X}^\top \mathbf{X} + \Sigma_p^{-1}\right)(\beta - \hat{\beta})\right\}$$

- Maximum A Posteriori (MAP) estimate of β :

$$\hat{\beta} = \frac{1}{\sigma_n^2} \left(\frac{1}{\sigma_n^2} \mathbf{X}^\top \mathbf{X} + \Sigma_p^{-1} \right)^{-1} \mathbf{X}^\top \mathbf{y}$$

Introduction to Gaussian Processes



Introduction to Gaussian Processes



Gaussian Process

$$f \sim \mathcal{GP}(m, k)$$

- Definition:

For any finite set $X = (x_1, \dots, x_n) \subset \mathcal{X}$ of any size $n \in \mathbb{N}$, the random vector

$$f_X = (f(x_1), \dots, f(x_n))^T \in \mathbb{R}^n$$

follows the multivariate normal distribution $N(m_X, k_{XX})$.

- $K(\cdot, \cdot)$ is positive definite.

Gaussian Process

- RBF kernel

$$k_{\gamma}(x, y) = \Phi_{\gamma}(x - y) = \exp(-\|x - y\|^2 / \gamma^2)$$

Gaussian Process

$$Y = f(\mathbf{x}) + \epsilon, \quad f|Y \sim \mathcal{GP}(\bar{m}, \bar{k}),$$

where $\bar{m}$ and $\bar{k}$ are given by

$$\begin{aligned}\bar{m}(x) &= m(x) + k_{xX}(k_{XX} + \sigma^2 I_n)^{-1}(Y - m_X) \\ \bar{k}(x, x') &= k(x, x') - k_{xX}(k_{XX} + \sigma^2 I_n)^{-1}k_{Xx'},\end{aligned}$$

where $k_{Xx} = k_{xX}^T = (k(x_1, x), \dots, k(x_n, x))^T$.

Introduction to Gaussian Processes



Introduction to Gaussian Processes



Kernel Ridge Regression (KRR)

$$\hat{f} = \arg \min_{f \in \mathcal{H}_k} \frac{1}{n} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}_k}^2$$



$$\hat{f}(x) = k_{xX}(k_{XX} + n\lambda I_n)^{-1}Y = \sum_{i=1}^n \alpha_i k(x, x_i), \quad x \in \mathcal{X},$$

where

$$(\alpha_1, \dots, \alpha_n)^T = (k_{XX} + n\lambda I_n)^{-1}Y \in \mathbb{R}^n.$$

Kernel Ridge Regression (KRR)

- KRR Predictive Function:

$$\hat{f}(x) = k_{xX}(k_{XX} + n\lambda I_n)^{-1}Y = \sum_{i=1}^n \alpha_i k(x, x_i), \quad x \in \mathcal{X},$$

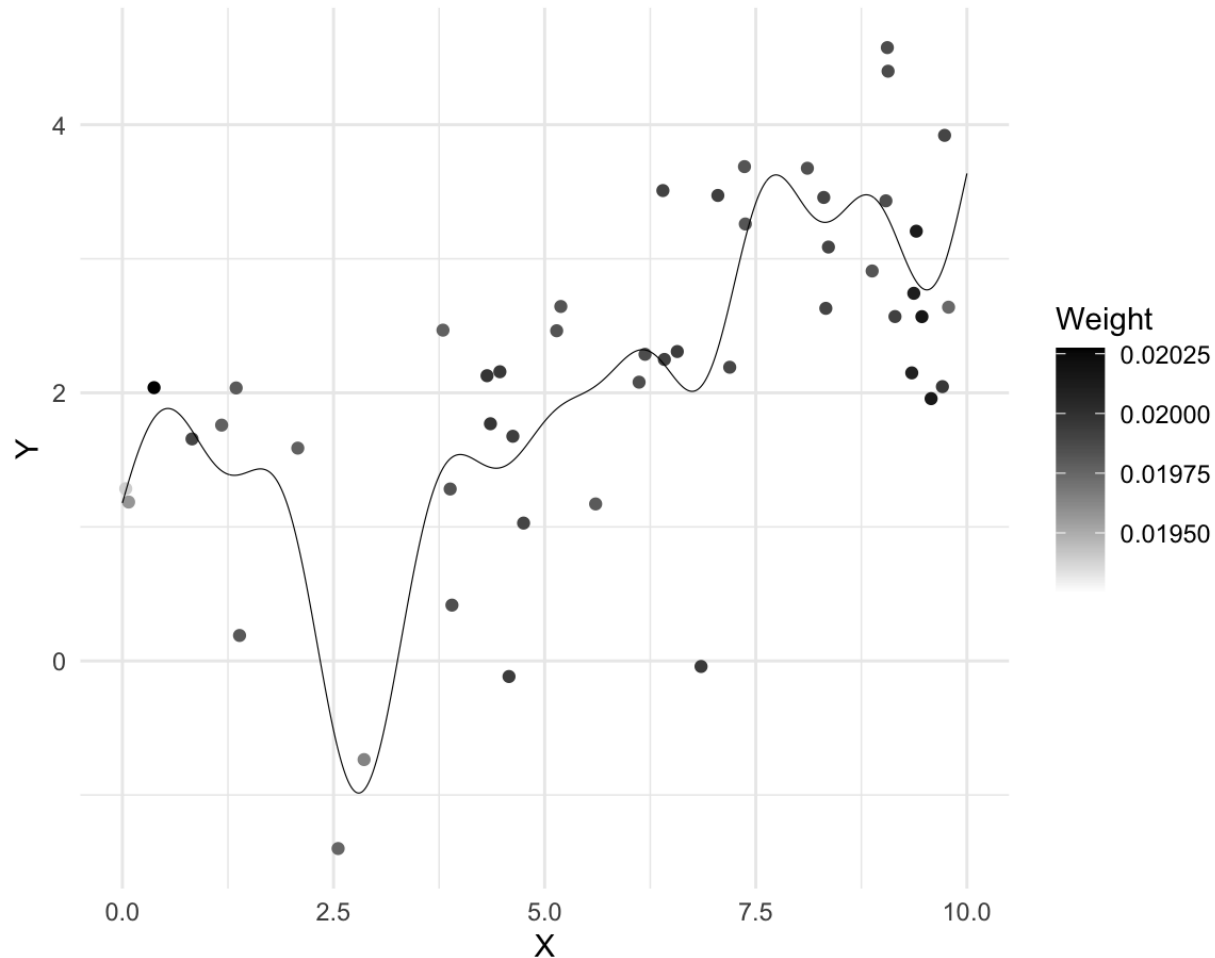
- Gaussian Process Predictive Mean:

$$\bar{m}(x) = m(x) + k_{xX}(k_{XX} + \sigma^2 I_n)^{-1}(Y - m_X)$$

- Condition:

$$1: \lambda = \sigma^2 \quad 2: m(x) = 0$$

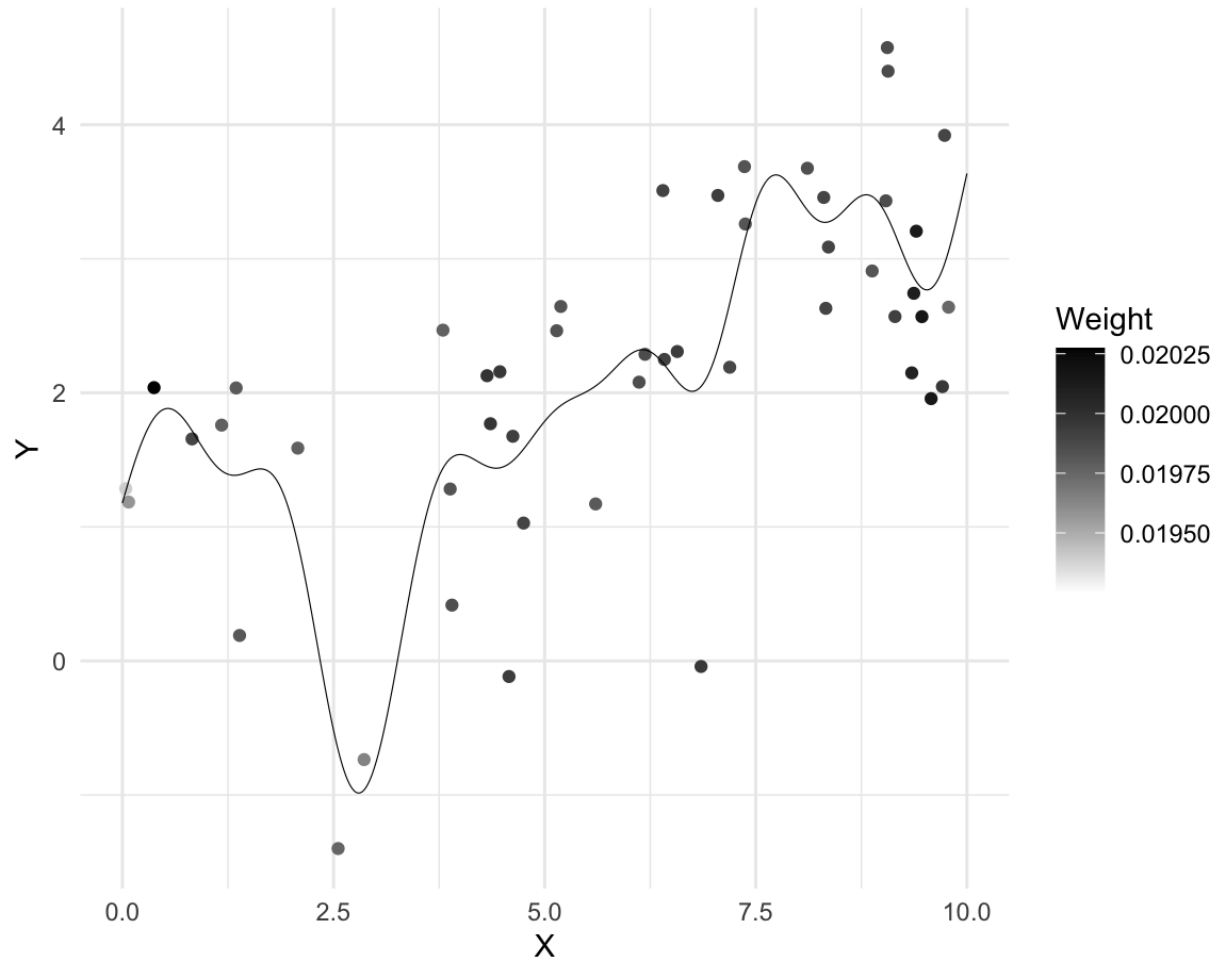
Kernel Ridge Regression (KRR)



Observation:

- $Y = 1 + 2\expit(2x - 10) + \epsilon$, where $\epsilon \sim N(0, 1)$
- The black smooth curve is the posterior mean function predicted by the KRR model.

Kernel Ridge Regression (KRR)



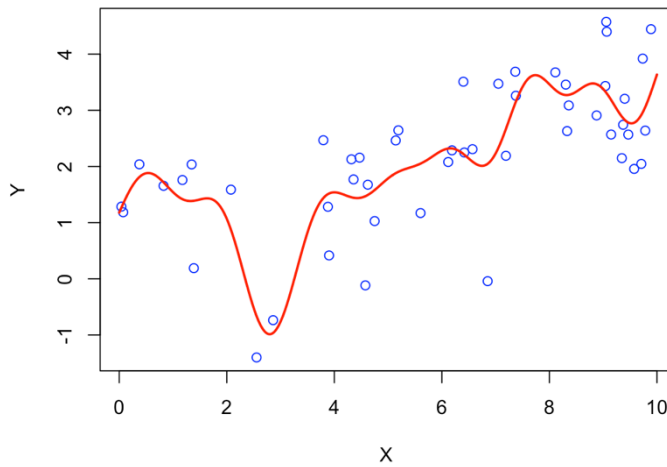
Observation:

- Color intensity of each data point corresponds to the relative weight of prediction on mean of y .
- $\rho=1$ (length scale), $\lambda=0.001$ (smoothness)

Kernel Ridge Regression (KRR)

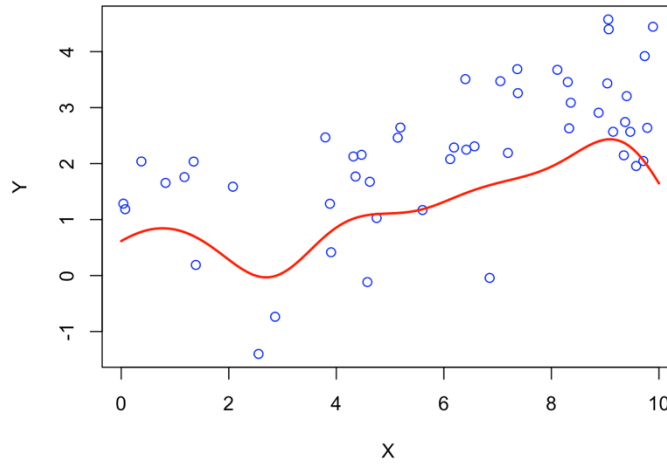
$\rho=1$:

Posterior Mean Function



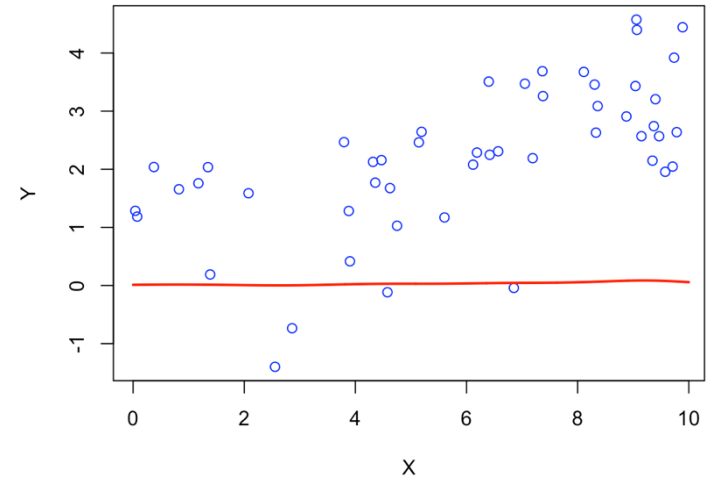
$\lambda = 0.001$

Posterior Mean Function



$\lambda = 0.1$

Posterior Mean Function



$\lambda = 10$

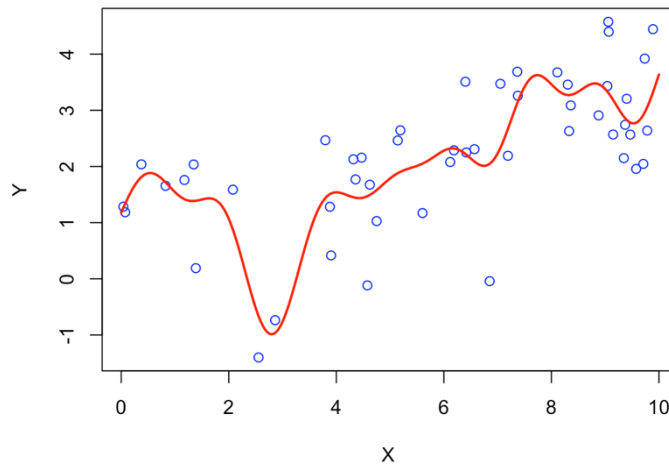
Magnitude & Smoothness



Kernel Ridge Regression (KRR)

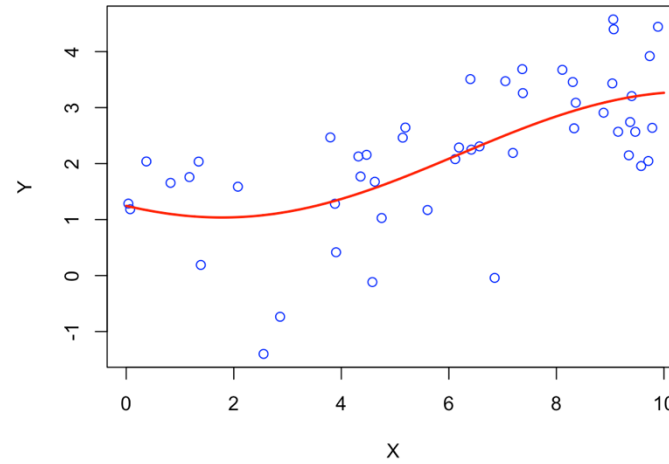
$\lambda=0.001$:

Posterior Mean Function



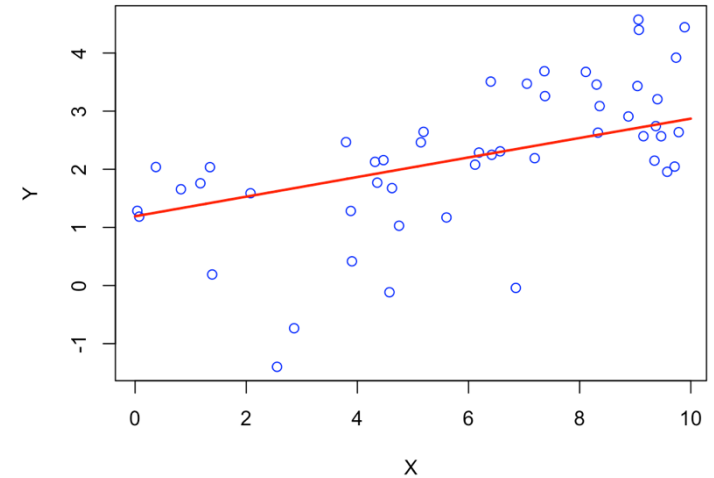
$\rho = 1$

Posterior Mean Function



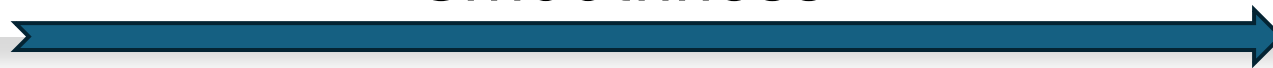
$\rho = 10$

Posterior Mean Function



$\rho = 100$

Smoothness





Thank you!





Q&A